Program / Source Code

```
#include <stdio.h>
int main()
{
    int previous, current, units;

    printf("Enter previous reading: ");
    scanf("%d", &previous);

    printf("Enter current reading: ");
    scanf("%d", &current);

    if (previous < 0 || current < 0 ||
        current < previous) {
        printf("Invalid meter reading\n");
    } else
{
        units = current - previous;
        printf("Units consumed = %d\n", units);
    }
return 0;
}
```

Output Screenshot



Program / Source Code

```
#include <stdio.h>
int main()
{
    long long previous, current, units;

    printf("Enter previous reading: ");
    scanf("%lld", &previous);

    printf("Enter current reading: ");
 scanf("%lld", &current);

    if (previous < 0 || current < 0 ||
        current < previous) {
        printf("Invalid reading or rollover detected\n");
    } else
{
        units = current - previous;
        printf("Units consumed = %lld\n", units)
 }
 return 0;
}
```

Output Screenshot



Program / Source Code

```
#include <stdio.h>
#define RATE1 1.50
#define RATE2 2.30
#define RATE3 3.00
int main()
 {
    int units;
    double charge;

    printf("Enter units consumed: ");
    scanf("%d", &units);

    if (units < 0) {
        printf("Invalid units\n");
    }
    else if (units <= 100) {
        charge = units * RATE1;
        printf("Energy charge = Rs. %.2f\n", charge);
    }
    else if (units <= 300) {
        charge = 100 * RATE1
```

```
+ (units - 100) * RATE2;

        printf("Energy charge = Rs. %.2f\n", charge);
    }
    else {
        charge = 100 * RATE1
                + 200 * RATE2
                + (units - 300) * RATE3;

        printf("Energy charge = Rs. %.2f\n", charge);
    }
```

Output

```
Enter units consumed: 150
Energy charge = Rs. 265.00
```

Program / Source Code:

```
#include <stdio.h>

static inline int square(int x)
{
    return x * x;
}

static inline int max(int a, int b)
{
    return (a > b) ? a : b;
}

int main()
{
    int a = 5, b = 10;

    printf("Square = %d\n", square(a));
    printf("Maximum = %d\n", max(a, b));

    return 0;
}
```

